

JAWAD ALI

AI Engineer

Rawalpindi/Islamabad, Pakistan | +92 315 5720296 | jawadaliraja2022@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

CAREER SUMMARY

Results-driven AI/ML Engineer and BS Artificial Intelligence graduate from COMSATS University Islamabad with hands-on experience building production-ready Generative AI systems, LLM applications, and end-to-end automation pipelines. Specializes in designing agentic workflows using LangChain and LangGraph, building Retrieval-Augmented Generation (RAG) pipelines with vector databases such as FAISS and Pinecone, and deploying LLM-powered chatbots and copilots via FastAPI and Streamlit. Strong practical experience with n8n for orchestrating multi-step automation workflows, integrating third-party APIs, and building event-driven data pipelines for real business operations. Proficient in MLOps practices including experiment tracking with MLflow, containerization with Docker, CI/CD pipelines, and cloud deployments on AWS, Azure, and GCP. Experienced in fine-tuning transformer models, prompt engineering, and integrating AI solutions with Copilot Studio for low-code agent orchestration. Solid background in classical and deep learning including CNNs, RNNs, LSTMs, and ensemble methods (XGBoost, LightGBM, CatBoost), with practical exposure across data preprocessing, model training, evaluation, and production deployment. Committed to delivering scalable, clean AI architectures that drive measurable business value.

EXPERIENCE

AI / ML Engineer

Aug 2025 – Present

OptimAlze BE (*Remote, Belgium*)

Responsibilities:

- Designed and deployed autonomous LangChain ReAct agents and LangGraph-based multi-agent systems, automating complex multi-step business workflows across cloud infrastructure.
- Built end-to-end AI automation workflows using n8n for intelligent scheduling, data ingestion, webhook handling, routing, and third-party API orchestration — enabling full business process automation.
- Implemented and deployed Retrieval-Augmented Generation (RAG) pipelines integrating FAISS and Pinecone vector databases for scalable semantic search and document-based question answering.
- Developed production-grade deep learning and classical ML pipelines for classification, regression, and NLP tasks; integrated trained models into live systems via FastAPI REST APIs.
- Connected and fine-tuned LLMs (OpenAI GPT-4, Claude API, HuggingFace Transformers) to CRMs, databases, and operational services, translating AI research into deployable automation solutions.
- Built interactive demo and evaluation interfaces using Streamlit to validate LLM outputs, RAG pipeline accuracy, and end-to-end agent workflows.
- Enforced MLOps best practices including MLflow experiment tracking, Docker containerization, and CI/CD pipelines — ensuring reproducible, scalable, cloud-ready AI systems on AWS and Azure.
- Engineered prompt engineering strategies and tool/function-calling architectures to improve LLM reasoning, reduce hallucinations, and enable reliable autonomous decision-making in agentic systems.
- Collaborated cross-functionally with product and engineering teams to deliver clean, maintainable AI architectures optimized for performance, scalability, and long-term reliability.

Tools & Technologies:

- Generative AI & Agents: LangChain, LangGraph, ReAct Agents, Agentic Workflows, Tool / Function Calling, Prompt Engineering, Copilot Studio
- LLMs & Models: OpenAI GPT-4 / GPT-4o, Claude API (Anthropic), HuggingFace Transformers, Sentence Transformers, Multilingual LLMs, LLM Fine-tuning
- Retrieval & Search: RAG Pipelines, Semantic Search, FAISS, Pinecone, Document Parsing (PDFs, Manuals), Vector Similarity Search
- MLOps & Deployment: Docker, MLflow, CI/CD Pipelines, AWS (EC2, S3, Lambda), Azure, GCP, Kubernetes, Environment Management, Logging & Monitoring

- Automation & Orchestration: n8n, Make.com-compatible Workflows, Webhook Handling, Scheduled Jobs, Event-Driven Pipelines, Third-Party API Integration
- Backend & APIs: Python, FastAPI, Flask, RESTful API Design, JWT Authentication, PostgreSQL, MongoDB, Redis, MySQL
- ML & Deep Learning: CNNs, RNNs, LSTMs, Transformers, Transfer Learning, scikit-learn, XGBoost, LightGBM, CatBoost, Ensemble Methods
- Data & NLP: Pandas, NumPy, NLTK, spaCy, Data Preprocessing, Feature Engineering, Statistical Analysis
- Frontend & Demos: Streamlit, React.js, Node.js, Express.js, MERN Stack
- Version Control & DevOps: Git, GitHub, CI/CD Fundamentals, Branching & Merging, Code Reviews

EDUCATION

COMSATS University Islamabad

2022 – 2026

BS in Artificial Intelligence

Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Science, Statistical Modelling, Neural Networks, Databases, Software Engineering

PROJECTS

Business Analyser Agent — Autonomous Generative AI System (2026)

Stack: Python • LangChain ReAct • OpenAI API • FastAPI • PostgreSQL • n8n • Docker • MLflow • AWS

- Built a multi-step autonomous agent using LangChain ReAct for dynamic business data analysis with tool-calling, memory management, and intelligent decision routing — deployed on AWS cloud infrastructure.
- Integrated n8n for automated scheduling, data ingestion pipelines, and cross-service orchestration, forming a complete AI-to-ops automation system aligned with real business workflows.
- Delivered FastAPI REST backend connecting the agent to PostgreSQL and third-party services; containerized with Docker and tracked with MLflow — demonstrating full production-grade MLOps delivery.

HEZOP Analyzer Agent (2025)

Stack: Python • LangChain • OpenAI API • AutoCAD Integration • FastAPI • Streamlit

- Built an AI-powered HEZOP analysis agent integrated with AutoCAD to analyze P&ID diagrams, identify process hazards, and assist HEZOP engineers with domain-specific safety queries.
- Implemented a RAG-based knowledge pipeline over engineering safety documents, enabling the agent to reason over technical standards and autonomously support risk assessment workflows.

Car Price Analyst Agent (2025)

Stack: Python • LangChain • OpenAI API • Selenium • BeautifulSoup • FastAPI • SMTP

- Developed an intelligent agent that scrapes second-hand car listings from multiple Belgium-based platforms, applies user-defined filters and preferences, and predicts fair market prices using ML models.
- Automated end-to-end pipeline from data scraping and price evaluation to autonomous email delivery — notifying users with recommendations and shortlisted vehicles matching their criteria.

Vehicle Insurance Prediction — MLOps Pipeline (2026)

Stack: Python • scikit-learn • XGBoost • MLflow • Docker • AWS S3

- Built a complete end-to-end machine learning pipeline for vehicle insurance claim prediction, covering data preprocessing, feature engineering, model training, and evaluation using scikit-learn and XGBoost.
- Implemented full MLOps lifecycle: versioned all experiments and model artifacts with MLflow Model Registry, containerized the inference service with Docker, and configured AWS S3 for artifact storage and cloud-ready deployment.
- Demonstrated production-grade ML engineering standards — reproducible pipelines, environment-isolated containers, and structured model versioning — applicable to real-world insurance risk analytics use cases.

SmartInfo — YouTube & Document Content Analyzer (Final Year Project, 2025)

Stack: Python • LangChain • HuggingFace Transformers • FAISS • OpenAI API • React.js • FastAPI • Streamlit

- Designed and implemented an end-to-end RAG pipeline leveraging FAISS vector search, sentence embeddings, and LangChain agent tools enabling intelligent Q&A over YouTube transcripts and uploaded documents.

- Fine-tuned HuggingFace transformer models for domain-specific NLP tasks; integrated React.js frontend and Streamlit demo interface with FastAPI backend for a seamless AI copilot experience.
- Significantly improved answer relevance and reduced hallucination through hybrid retrieval strategies and advanced prompt engineering, optimizing for cost-efficiency and response quality.

SKILLS

- Professional: LangChain, LangGraph, Agentic AI, Chatbots, LLMs / Generative AI, FastAPI, Flask, JavaScript (Node.js / Express), React.js
- Automation Tools: Copilot Studio, n8n, Make.com, Webhook Orchestration, Event-Driven Workflows
- AI Models & APIs: OpenAI API, Claude API (Anthropic), HuggingFace Transformers, Google AI / Gemini, Mistral AI, Ollama
- Vector Databases & Search: FAISS, Pinecone, ChromaDB, Elasticsearch, OpenSearch
- Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, SQL Server, Firebase
- MLOps & DevOps: MLflow, Docker, Kubernetes, CI/CD Pipelines, AWS (EC2, S3, Lambda), Azure, GCP, Git, GitHub
- ML & Deep Learning: scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow / Keras, PyTorch, Transfer Learning
- Data & NLP: Pandas, NumPy, NLTK, spaCy, Selenium, BeautifulSoup, Data Preprocessing, Feature Engineering
- Frontend & Demos: Streamlit, Gradio, React.js, MERN Stack

CERTIFICATIONS

Anthropic AI Courses (May 2026)

Completed: Claude 101, Claude Code In Action, Claude With Anthropic API, Intro To Agent Skills, Intro to MCP